

III Semester BCA – Computer Networks

Unit – I : Full Answers

1. A computer network is a collection of interconnected computers and devices that share data and resources.
2. VPN – Virtual Private Network; RFID – Radio Frequency Identification.
3. GPS – Global Positioning System; NFC – Near Field Communication.
4. DMCA – Digital Millennium Copyright Act; CAPTCHA – Completely Automated Public Turing test to tell Computers and Humans Apart.
5. WiMAX – Worldwide Interoperability for Microwave Access; NSP – Network Service Provider.
6. ARPANET – Advanced Research Projects Agency Network; ICMP – Internet Control Message Protocol.
7. SMTP – Simple Mail Transfer Protocol; RTP – Real-time Transport Protocol.
8. HTTP – HyperText Transfer Protocol; TELNET – Telecommunication Network.
9. FTP – File Transfer Protocol; DNS – Domain Name System.
10. TCP – Transmission Control Protocol; UDP – User Datagram Protocol.
11. DSL – Digital Subscriber Line; SONET – Synchronous Optical Network.
12. Peer-to-peer model: All systems act as both client and server without centralized control.
13. Client-server model: Clients request services and servers provide them.
14. Forms of e-commerce: B2B, B2C, C2C, C2B, G2C.
15. Profiling: Collecting user data; Phishing: Fraud to steal sensitive information.
16. Transmission technologies: Broadcast and Point-to-Point.
17. Unicasting: One-to-one; Broadcasting: One-to-many communication.
18. Communication subnet consists of routers and transmission lines.
19. Components of subnet: Transmission lines and routers.
20. Routing decides best path; Forwarding moves packets to next hop.
21. Addressing identifies devices; Flow control manages data rate.
22. Store-and-forward waits for full packet; Cut-through forwards immediately.
23. Service primitives: Listen, Connect, Accept, Send, Receive, Disconnect.
24. Service defines what a layer does; Protocol defines how it is done.
25. OSI has 7 layers; TCP/IP has 4 layers.
26. Uses of networks: Resource sharing, communication, information sharing, cost reduction, reliability, scalability.